IMO 2026 — Problem 2

Solution by GPT-5.6 Sol

2026-07-17

Source: `results/imo-2026/gpt-5.6-sol/problem-02/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let ABC be a triangle and let points M and N be the midpoints of sides AB and AC, respectively. Let points K and L be chosen inside triangles BMC and BNC, respectively, such that K lies inside the angle LBA, L lies inside the angle ACK, and $\angle KBA = \angle ACL$, $\angle LBK = \angle LNC$, $\angle LCK = \angle BMK$. Let O be the circumcentre of triangle AKL. Prove that $OM = ON$.

Solution document

Status

solved

Approaches tried

- Oblique coordinates based at A, with the circle represented by its two linear coefficients — all branch, ray-intersection, and circle-system issues were settled, reducing the theorem to one explicit determinant identity; the requested final human-checkable cancellation remained unproved in round 1.
- Quadratic ray-length elimination followed by an explicit low-degree certificate — worked: the two closure conditions give independent quadratics in BK and CL, and a coefficient-by-coefficient audit proves that their displayed linear combination is the required determinant identity.

Current best

The oblique-coordinate approach is complete. After the certified oriented-ray reconstruction, the two closure conditions yield quadratics $Q_t = Q_s = 0$. The determinant expression equivalent to $OM = ON$ is an explicit linear combination of these quadratics; the full coefficient audit is included below.

Full proof

Let

$$\alpha = \angle BAC, \quad \beta = \angle ABC, \quad \gamma = \angle BCA, \qquad a = BC, \quad b = CA, \quad c = AB.$$

Set

$$x = \angle KBA = \angle ACL, \quad y = \angle LBK = \angle LNC, \quad z = \angle LCK = \angle BMK.$$

Reflecting the configuration if necessary, which preserves angles and distances, put

$$A = (0,0), \quad B = (c,0), \quad C = (b\cos\alpha, b\sin\alpha)$$

with the oriented determinant $[B, C] > 0$.

We first invoke the certified **oriented positive-ray reconstruction lemma**. The strict ray orders forced by the interiority hypotheses give

$$0 < x < \min(\beta,\gamma), \qquad 0 < y < \beta - x, \qquad 0 < z < \gamma - x. \tag{1}$$

Writing $e(\theta) = (\cos\theta, \sin\theta)$, intersection of the positive rays from B and the midpoint $M = B/2$, and similarly from C and $N = C/2$, gives

$$K = B - te(-x), \qquad t = BK = \frac{c\sin z}{2\sin(x+z)}, \tag{2}$$

$$L = C - se(\alpha + x), \qquad s = CL = \frac{b\sin y}{2\sin(x+y)}. \tag{3}$$

For completeness, (2) follows from $B - te(-x) = B/2 + ue(z)$ by taking determinants with the two direction vectors; the calculation at C is identical with the indicated directions. By (1), all the sines in (2) and (3) are positive, so $s, t > 0$, and no reflected or negative-ray branch has been introduced.

The same lemma, using the **Sine Law** (Knowledge Base, Synthetic toolkit) in triangles BKC and BLC, gives the remaining closure conditions

$$t = a\frac{\sin(\gamma - x - z)}{\sin(\alpha + 2x + z)}, \tag{4}$$

$$s = a\frac{\sin(\beta - x - y)}{\sin(\alpha + 2x + y)}. \tag{5}$$

The denominators are positive because they are sines of angles of the respective nondegenerate triangles.

We eliminate z from (2) and (4). Equation (2), expanded, gives

$$\cot z = \frac{c - 2t\cos x}{2t\sin x}. \tag{6}$$

This division is legitimate because $t, \sin x, \sin z > 0$. Cross-multiplying (4), expanding the sines containing z, and dividing by $\sin z$, we get

$$t\{\sin(\alpha + 2x)\cot z + \cos(\alpha + 2x)\} = a\{\sin(\gamma - x)\cot z - \cos(\gamma - x)\}. \tag{7}$$

Substitute (6), multiply by $2t\sin x$, and move the right side left:

$$\begin{aligned}0 &= \{t\sin(\alpha+2x) - a\sin(\gamma-x)\}(c-2t\cos x)\\ &\quad + 2t\sin x\{t\cos(\alpha+2x) + a\cos(\gamma-x)\}.\end{aligned}$$

Using $\sin U\cos x - \cos U\sin x = \sin(U-x)$, this becomes

$$0 = -ac\sin(\gamma-x) + 2at\sin\gamma + ct\sin(\alpha+2x) - 2t^2\sin(\alpha+x). \tag{8}$$

The Sine Law in ABC gives $a\sin\gamma = c\sin\alpha$. Also

$$a\sin(\gamma-x) = c\sin(\alpha+x) - b\sin x. \tag{9}$$

Indeed, expand the left side and use $a\sin\gamma = c\sin\alpha$ and the cosine-law projection $a\cos\gamma = b - c\cos\alpha$. Substitution in (8) proves

$$Q_t := bc\sin x - c^2\sin(\alpha+x) + ct\{2\sin\alpha + \sin(\alpha+2x)\} - 2t^2\sin(\alpha+x) = 0. \tag{10}$$

The elimination of y is analogous but is written out. Equations (3) and (5) give

$$\cot y = \frac{b-2s\cos x}{2s\sin x}$$

and

$$s\{\sin(\alpha+2x)\cot y + \cos(\alpha+2x)\} = a\{\sin(\beta-x)\cot y - \cos(\beta-x)\}.$$

Division is valid because $s, \sin x, \sin y > 0$. Substitution and multiplication by $2s\sin x$ gives

$$0 = -ab\sin(\beta-x) + 2as\sin\beta + bs\sin(\alpha+2x) - 2s^2\sin(\alpha+x). \tag{11}$$

The Sine Law gives $a\sin\beta = b\sin\alpha$, and the projection identity $a\cos\beta = c - b\cos\alpha$ gives $a\sin(\beta-x) = b\sin(\alpha+x) - c\sin x$. Thus

$$Q_s := bc\sin x - b^2\sin(\alpha+x) + bs\{2\sin\alpha + \sin(\alpha+2x)\} - 2s^2\sin(\alpha+x) = 0. \tag{12}$$

We now invoke the certified **oblique circle-coefficient criterion**, which uses coordinates, Cramer's rule, and the **Power of a Point identity** (Knowledge Base, Synthetic toolkit). For $d = C - B$, it states that the circle through A, K, L satisfies $OM = ON$ if and only if

$$F := 2(|K|^2[L,d] - |L|^2[K,d]) - (c^2-b^2)[K,L] = 0. \tag{13}$$

There is no singular case: the stated circumcentre of triangle AKL presupposes that A, K, L are noncollinear, so $[K,L] \neq 0$, exactly the criterion's nonsingularity condition. Briefly, the criterion writes the circle as $|X|^2 - U\cdot X = 0$; equality of its powers at $B/2, C/2$ is equivalent to a linear relation between the two oblique coefficients, and Cramer's rule converts that relation into (13).

It remains to prove (13). Abbreviate

$$p = \sin(\alpha+x), \quad q = \sin x, \quad r = \sin\alpha, \quad u = \sin(\alpha-x), \quad h = 2r + \sin(\alpha+2x).$$

From (2) and (3), scalar-product and determinant calculations give

$$|K|^2 = c^2 - 2ct\cos x + t^2, \quad |L|^2 = b^2 - 2bs\cos x + s^2, \tag{14}$$

$$[K,L] = bcr - csp - btp + ts\sin(\alpha+2x), \qquad (15)$$

$$[L,d] = bcr + s(bq-cp), \quad [K,d] = bcr + t(-bp+cq). \qquad (16)$$

Multiplying (14)–(16) in (13) and collecting gives the complete coefficient table

monomial	coefficient in F
1	$bc(c^2-b^2)r$
t	$b^3p - 2b^2cq - 2bc^2u - bc^2p$
s	$2b^2cu + b^2cp + 2bc^2q - c^3p$
t^2	$2bcr$
ts	$(c^2-b^2)h$
s^2	$-2bcr$
t^2s	$2(bq-cp)$
ts^2	$2(bp-cq).$

(17)

No other monomial can occur by (14)–(16). The entries in (17) use only the sine addition formulas; in particular the ts-coefficient simplifies to $(c^2-b^2)h$.

We claim the explicit certificate

$$\boxed{\begin{aligned} 2pF = &-\big(2bcr + 2s(bq-cp)\big)Q_t \\ &-\big(-2bcr + 2t(bp-cq)\big)Q_s. \end{aligned}} \qquad (18)$$

Let R denote its right side. Multiplying its two linear factors by (10) and (12), and comparing with twice p times (17), gives this full audit:

	$[2pF]$	$[R]$
1	$2bcpr(c^2-b^2)$	$2bcpr(c^2-b^2)$
t	$2b(b^2p^2 - 2bcpq - c^2p^2 - 2c^2pu)$	$2b(b^2p^2 - 2bcpq - c^2hr + c^2q^2)$
s	$2c(b^2p^2 + 2b^2pu + 2bcpq - c^2p^2)$	$2c(b^2hr - b^2q^2 + 2bcpq - c^2p^2)$
t^2	$4bcpr$	$4bcpr$
ts	$2p(c^2-b^2)h$	$2p(c^2-b^2)h$
s^2	$-4bcpr$	$-4bcpr$
t^2s	$4p(bq-cp)$	$4p(bq-cp)$
ts^2	$4p(bp-cq)$	$4p(bp-cq).$

(19)

The t- and s-rows agree by

$$hr - q^2 = p^2 + 2pu. \qquad (20)$$

To derive (20) by product-to-sum,

$$hr = 1 - \cos 2\alpha + \frac{\cos 2x - \cos(2\alpha+2x)}{2},$$

while

$$q^2 + p^2 + 2pu = 1 - \cos 2\alpha + \frac{\cos 2x - \cos(2\alpha+2x)}{2}.$$

Thus every row of (19) agrees, proving (18) by equality of polynomial coefficients.

Finally, (10) and (12) give $Q_t = Q_s = 0$, so (18) yields $2pF = 0$. Since $x < \beta$ and $\alpha+\beta<\pi$, we have $0 < \alpha + x < \pi$, hence $p > 0$. Therefore $F = 0$. By the oblique circle-coefficient criterion, $OM = ON$, as required.